Mikhail Razakov

Compiler Engineer · LLVM/C++ · Program Analysis

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I build compilers and program-analysis tools in C++ and .NET. At MCST I wrote an LLVM LICM pass and a test harness that compares behavior before and after optimization. In UniversalToolchain/Wist2 I work on SSA, the interpreter, and the CIL backend. A talk about the project was accepted for LangDev 2026.

LLVM LICM pass
C++ · LoopInfo · before/after optimization tests

#1 of 49 · 104/104
PS-form Memory Dependence Analyzer

LangDev 2026
UniversalToolchain/Wist2 talk accepted

EXPERIENCE

July 1 — August 31, 2026

MCST — compiler engineering intern

- Wrote an LLVM LICM pass in C++, checking expression invariance, preheaders, and whether instructions can be moved safely around side effects and memory accesses.
- Added a test harness with baseline/optimized execution through lli, opt -verify-each, and separate cases where the pass must or must not change the IR.

2026

PS-form Memory Dependence Analyzer — #1 of 49

- Implemented analysis of whether parametric memory accesses can overlap; passed 104/104 official tests and ranked #1 of 49.
- Used exact enumeration on small domains plus randomized/metamorphic tests to produce reproducible counterexamples.

2024 — present

UniversalToolchain / Wist2 — creator and primary developer

- Building a compiler/runtime platform with Bytecode/AIR, SSA, an interpreter, and a CIL backend.
- Check interpreter/CIL behavior parity and transformation correctness with tests and verifiers.

PROJECTS

UniversalToolchain / Wist2

The current exact test manifest passes with zero failures; interpreter/CIL parity, clean-consumer scenarios, and reproducible compiler/runtime contracts are verified.

PS-form Memory Dependence Analyzer

Ranked 1st of 49 with 104/104 tests, with reproducible wrong-answer and time-limit counterexamples.

x86-64 Codegen & Register Allocation Lab

A measurable pipeline with spill/reload metrics, code size, and reproducible result comparison.

SKILLS

Compiler / LLVM
C++23, LLVM passes, LoopInfo, CFG, LICM, SSA, dominance

Program analysis
memory dependence, dataflow, exact oracle on small domains, differential/randomized testing

IR / runtime
.NET, Bytecode/AIR, SSA, interpreter, CIL backend, verifier

Tooling
Python, CMake, Linux, clang-tidy, ASan/UBSan, Git

EDUCATION

Software Engineering student at HSE University

RECOGNITION

The LangDev 2026 Program Committee accepted a talk on the architecture of UniversalToolchain/Wist2 for presentation in Torremolinos, Spain.

Moscow / remote